

# SOUVIK BISWAS

Full-Stack Developer · New Grad 2026

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## SUMMARY

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Final-year Computer Science student and full-stack developer experienced in building production-grade web applications with the MERN stack, Next.js, and Go. Shipped a live multi-tenant platform serving 1,000+ users, with hands-on experience in AI integration, Docker, and cloud deployment.

## SKILLS

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**Languages:** C++, C#, JavaScript, TypeScript, Go

**Web & App:** MERN Stack, Next.js, React, SvelteKit, React Native (Expo), Tailwind CSS

**Backend & Databases:** Node.js, Express, Go (Gin), GraphQL, WebSockets, MongoDB, Mongoose, PostgreSQL, Redis, BullMQ

**DevOps & Cloud:** AWS, Docker, Kubernetes, Docker Compose, Cloudflare Workers, Caddy Proxy

**AI & Integrations:** Google Gemini API, LangChain, OpenAI API, Meta WhatsApp Cloud API, Razorpay

**Core CS:** Data Structures and Algorithms (200+ problems solved), Low-Level Design, Distributed Systems, DBMS, Computer Networks, Cloud Computing, Cryptography and Network Security

## PROJECTS

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### School Portal · Multi-Tenant School ERP

*Next.js, React, MongoDB, Redis, Docker, Gemini API*

- Architected and shipped a multi-tenant web platform solo for a startup, now serving **1,000+ active students** with strict tenant isolation enforced at the ORM layer via AsyncLocalStorage and a custom Mongoose plugin.
- Built an AI-powered attendance pipeline using **Google Gemini** OCR that extracts roll-wise data from photographed registers and validates it against the class roster, reducing manual attendance entry by 100%.
- Engineered a reliable background-job system with **BullMQ and Redis** for Meta WhatsApp API push notifications and Dockerized cron jobs for idempotent per-student Razorpay billing.

### DriftWatch · Drift Detection SaaS

*Go (Gin), SvelteKit, Cloudflare Workers, Postgres, Redis, Gemini*

- Built a multi-user SaaS in **Go (Gin)** with a goroutine-based cron scheduler that diffs live Docker Engine state against the GitHub-declared `docker-compose.yml`, detecting **5 drift categories** (env, image, port, missing/extra container) within 60 seconds of any state change.
- Designed a secure agent-push architecture: a lightweight agent pushes container state out over HTTPS with per-project keys — no inbound access to user servers — with per-user GitHub tokens encrypted at rest using **AES-256-GCM**.
- Eliminated redundant LLM calls by gating Gemini fix-suggestions behind a **SHA256 state-hash dedupe** in Upstash Redis, and shipped a Cloudflare Worker for GitHub webhook ingestion with HMAC-SHA256 signature verification.

### ReviewFlow · Real-Time Code Review

*Svelte 5, Go (Gin), GraphQL, WebSocket, MongoDB, Gemini*

- Engineered a race-free WebSocket room hub in **Go**: an RWMutex-guarded room map mutated only inside a single `Run()` goroutine fed by register/unregister/broadcast channels, supporting hundreds of concurrent reviewers with bounded send buffers and slow-client eviction.
- Designed a **gqlgen GraphQL** layer bridged to the hub so every review mutation persists to MongoDB, fans out a typed frame to all connected viewers, and writes a notification row — live, room-scoped pub/sub with no separate message broker.
- Built an asynchronous **Gemini** code-review pipeline behind a 10 req/min rate limiter with strict JSON output, 30s timeout, retry on quota exhaustion, and idempotent MongoDB upserts, broadcasting results to the room the instant the model returns.

### PrepSenseAI · AI Interview Prep Platform

*React, Express, Gemini API, MongoDB, Puppeteer, Zod*

- Built a full-stack platform that parses a user's PDF resume against a target job description to generate a personalized interview preparation strategy.
- Integrated the Google Gemini SDK with **Zod** schema validation to enforce strict JSON outputs, generating technical and behavioral questions with a color-coded skill-gap analysis.
- Developed a server-side HTML-to-PDF rendering pipeline with **Puppeteer** to auto-generate ATS-optimized resumes from AI feedback, cutting resume turnaround to under 30 seconds.

## EDUCATION

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### Siliguri Institute of Technology

Bachelor of Technology in Computer Science and Engineering · CGPA: 7.5/10

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